Partner Checks: Teacher Answer Key

Each row lists Partner A's problem, Partner B's problem and the shared answer. Every pair was solved independently from its printed text by the build script (tools/discrete-math/partner). Answer conventions: sets in roster form with elements in increasing order, ∅ for the empty set; truth columns as T/F strings in standard row order (TT, TF, FT, FF; with r: TTT, TTF, ..., FFF); fractions in lowest terms; ordered pairs as (x, y).

Unit 1 · Speaking Mathematically · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	Domain D = {−3, −2, ..., 3}. Find the truth set of "2n + 1 > 4".	Domain D = {−3, −2, ..., 3}. Find the truth set of "n ≥ 2".	**{2, 3}**
2	A has 2 elements and B has 5 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 2 letters followed by one digit from a set of 5 digits. How many codes are possible?	**10**
3	How many elements are in {n ∈ ℤ | 4 ≤ n ≤ 8}?	How many positive multiples of 3 are at most 15?	**5**
4	How many subsets does a set with 3 elements have (include ∅)?	How many bit strings of length 3 are there?	**8**
5	Solve (3x − 3, y + 2) = (9, 3). Give (x, y).	Solve (x + y, x − y) = (5, 3). Give (x, y).	**(4, 1)**
6	A has 3 elements and B has 2 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 3 letters followed by one digit from a set of 2 digits. How many codes are possible?	**6**

Unit 1 · Speaking Mathematically · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	A has 5 elements and B has 7 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 7 letters followed by one digit from a set of 5 digits. How many codes are possible?	**35**
2	Solve (2x − 2, y + 2) = (8, 6). Give (x, y).	Solve (x + y, x − y) = (9, 1). Give (x, y).	**(5, 4)**
3	A has 6 elements and B has 7 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 7 letters followed by one digit from a set of 6 digits. How many codes are possible?	**42**
4	How many subsets does a set with 3 elements have (include ∅)?	How many bit strings of length 3 are there?	**8**
5	Domain D = {−3, −2, ..., 3}. Find the truth set of "n² − n = 0".	Domain D = {−3, −2, ..., 3}. Find the truth set of "n(n − 1) = 0".	**{0, 1}**
6	How many elements are in {n ∈ ℤ | -2 ≤ n ≤ 7}?	How many positive multiples of 3 are at most 30?	**10**

Unit 1 · Speaking Mathematically · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	Solve (3x − 3, y + 2) = (3, 7). Give (x, y).	Solve (x + y, x − y) = (7, -3). Give (x, y).	**(2, 5)**
2	Domain D = {−3, −2, ..., 3}. Find the truth set of "n(n − 1) = 0".	Domain D = {−3, −2, ..., 3}. Find the truth set of "$n^2 - n = 0$".	**{0, 1}**
3	A has 7 elements and B has 2 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 7 letters followed by one digit from a set of 2 digits. How many codes are possible?	**14**
4	How many subsets does a set with 7 elements have (include ∅)?	How many bit strings of length 7 are there?	**128**
5	How many elements are in {n ∈ ℤ | -2 ≤ n ≤ 5}?	How many positive multiples of 3 are at most 24?	**8**
6	A has 7 elements and B has 5 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 7 letters followed by one digit from a set of 5 digits. How many codes are possible?	**35**

Unit 1 · Speaking Mathematically · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	Solve (3x − 3, y + 2) = (12, 8). Give (x, y).	Solve (x + y, x − y) = (11, -1). Give (x, y).	**(5, 6)**
2	How many subsets does a set with 3 elements have (include ∅)?	How many bit strings of length 3 are there?	**8**
3	Domain D = {−3, −2, ..., 3}. Find the truth set of "$n^3 = n$".	Domain D = {−3, −2, ..., 3}. Find the truth set of "$n^2 = 1$ or $n = 0$".	**{-1, 0, 1}**
4	A has 2 elements and B has 5 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 2 letters followed by one digit from a set of 5 digits. How many codes are possible?	**10**
5	How many subsets does a set with 2 elements have (include ∅)?	How many bit strings of length 2 are there?	**4**
6	How many elements are in {n ∈ ℤ | -4 ≤ n ≤ 2}?	How many positive multiples of 3 are at most 21?	**7**

Unit 1 · Speaking Mathematically · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	Domain D = {−3, −2, ..., 3}. Find the truth set of "n ≥ 2".	Domain D = {−3, −2, ..., 3}. Find the truth set of "n + 2 > 3".	**{2, 3}**
2	How many subsets does a set with 3 elements have (include ∅)?	How many bit strings of length 3 are there?	**8**
3	Solve (2x − 2, y + 2) = (4, 6). Give (x, y).	Solve (x + y, x − y) = (7, -1). Give (x, y).	**(3, 4)**
4	How many elements are in {n ∈ ℤ | 0 ≤ n ≤ 2}?	How many positive multiples of 3 are at most 9?	**3**
5	A has 6 elements and B has 6 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 6 letters followed by one digit from a set of 6 digits. How many codes are possible?	**36**
6	How many subsets does a set with 4 elements have (include ∅)?	How many bit strings of length 4 are there?	**16**

Unit 1 · Speaking Mathematically · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Domain D = {−3, −2, ..., 3}. Find the truth set of "n + 2 > 3".	Domain D = {−3, −2, ..., 3}. Find the truth set of "n ≥ 2".	**{2, 3}**
2	A has 2 elements and B has 4 elements. How many ordered pairs are in A × B?	A code is one letter from a set of 2 letters followed by one digit from a set of 4 digits. How many codes are possible?	**8**
3	How many elements are in {n ∈ ℤ | -3 ≤ n ≤ -1}?	How many positive multiples of 3 are at most 9?	**3**
4	How many subsets does a set with 5 elements have (include ∅)?	How many bit strings of length 5 are there?	**32**
5	How many subsets does a set with 8 elements have (include ∅)?	How many bit strings of length 8 are there?	**256**
6	Solve (3x − 3, y + 2) = (0, 6). Give (x, y).	Solve (x + y, x − y) = (5, -3). Give (x, y).	**(1, 4)**

Unit 2 · The Logic of Compound Statements · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	Give the truth column of ¬p → ¬q (standard row order, T/F string).	Give the truth column of p ∨ ¬q (standard row order, T/F string).	**TTFT**
2	Give the truth column of ¬(¬p ∨ ¬q) (standard row order, T/F string).	Give the truth column of p ∧ q (standard row order, T/F string).	**TFFF**
3	How many rows does a truth table with 3 variables have?	How many rows does the truth table for p ∧ q ∧ r have?	**8**
4	In how many rows of its truth table (variables p, q) is p ∧ (p ∨ q) true?	In how many rows of its truth table (variables p, q) is p ⊕ q true?	**2**
5	In how many rows of its truth table (variables p, q) is q → p true?	In how many rows of its truth table (variables p, q) is ¬(p ∧ ¬q) true?	**3**
6	How many rows does a truth table with 4 variables have?	How many rows does the truth table for p ∧ q ∧ r ∧ s have?	**16**

Unit 2 · The Logic of Compound Statements · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	Premises: p ∨ q; q. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; q. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = T)**
2	Give the truth column of (p → q) ∧ (q → p) (standard row order, T/F string).	Give the truth column of ¬(p ⊕ q) (standard row order, T/F string).	**TFFT**
3	Premises: p ↔ q; r. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; r. Conclusion: q. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = F, r = T)**
4	In how many rows of its truth table (variables p, q) is p ∧ q true?	In how many rows of its truth table (variables p, q) is ¬(¬p ∨ ¬q) true?	**1**
5	In how many rows of its truth table (variables p, q) is ¬p → ¬q true?	In how many rows of its truth table (variables p, q) is ¬p → q true?	**3**
6	Give the truth column of ¬p → q (standard row order, T/F string).	Give the truth column of p ∨ q (standard row order, T/F string).	**TTTF**

Unit 2 · The Logic of Compound Statements · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	Premises: ¬q → ¬p; ¬p. Conclusion: ¬q. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; ¬p. Conclusion: ¬q. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = T)**
2	In how many rows of its truth table (variables p, q, r) is (p ∨ q) → r true?	In how many rows of its truth table (variables p, q, r) is (p → r) ∧ (q → r) true?	**5**
3	Give the truth column of (p ∨ q) ∧ (p ∨ r) (standard row order, T/F string).	Give the truth column of p ∨ (q ∧ r) (standard row order, T/F string).	**TTTTTFFF**
4	In how many rows of its truth table (variables p, q) is ¬(p ∨ q) true?	In how many rows of its truth table (variables p, q) is p ∧ q true?	**1**
5	Premises: p ↔ q; r. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; r. Conclusion: q. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = F, r = T)**
6	Give the truth column of (p ∧ q) ∨ (p ∧ r) (standard row order, T/F string).	Give the truth column of p ∧ (q ∨ r) (standard row order, T/F string).	**TTTFFFFF**

Unit 2 · The Logic of Compound Statements · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	In how many rows of its truth table (variables p, q) is ¬(p ∧ ¬q) true?	In how many rows of its truth table (variables p, q) is p ∨ ¬q true?	**3**
2	Give the truth column of ¬q → ¬p (standard row order, T/F string).	Give the truth column of ¬(p ∧ ¬q) (standard row order, T/F string).	**TFTT**
3	In how many rows of its truth table (variables p, q) is p ∧ q true?	In how many rows of its truth table (variables p, q) is ¬(p ∨ q) true?	**1**
4	How many rows does a truth table with 3 variables have?	How many rows does the truth table for p ∧ q ∧ r have?	**8**
5	How many rows does a truth table with 5 variables have?	How many rows does the truth table for p ∧ q ∧ r ∧ s ∧ t have?	**32**
6	Give the truth column of ¬(¬p ∨ q) (standard row order, T/F string).	Give the truth column of ¬(p → q) (standard row order, T/F string).	**FTFF**

Unit 2 · The Logic of Compound Statements · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	Give the truth column of ¬(p ∨ q) (standard row order, T/F string).	Give the truth column of ¬p ∧ ¬q (standard row order, T/F string).	**FFFT**
2	In how many rows of its truth table (variables p, q) is ¬(p → q) true?	In how many rows of its truth table (variables p, q) is ¬(p ∨ q) true?	**1**
3	Premises: p ↔ q; r. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; r. Conclusion: q. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = F, r = T)**
4	Premises: p → q; q. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: p → q; ¬p. Conclusion: ¬q. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = T)**
5	Give the truth column of p ∧ q (standard row order, T/F string).	Give the truth column of ¬(¬p ∨ ¬q) (standard row order, T/F string).	**TFFF**
6	In how many rows of its truth table (variables p, q, r) is (p ∧ q) → r true?	In how many rows of its truth table (variables p, q, r) is p → (q → r) true?	**7**

Unit 2 · The Logic of Compound Statements · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Give the truth column of (p ∧ q) → r (standard row order, T/F string).	Give the truth column of p → (q → r) (standard row order, T/F string).	**TFTTTTTT**
2	Premises: p ∨ q; q. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	Premises: ¬p ∨ q; q. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = T)**
3	Give the truth column of (p ∧ q) ∨ (p ∧ r) (standard row order, T/F string).	Give the truth column of p ∧ (q ∨ r) (standard row order, T/F string).	**TTTFFFFF**
4	In how many rows of its truth table (variables p, q) is ¬p → q true?	In how many rows of its truth table (variables p, q) is p ∨ ¬q true?	**3**
5	Premises: p → q; r. Conclusion: q. Give the only row that makes every premise true and the conclusion false.	Premises: p ↔ q; r. Conclusion: p. Give the only row that makes every premise true and the conclusion false.	**(p = F, q = F, r = T)**
6	In how many rows of its truth table (variables p, q) is p ∧ ¬q true?	In how many rows of its truth table (variables p, q) is ¬(¬p ∨ q) true?	**1**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	Find gcd(6, 30).	Find gcd(48, 54).	**6**
2	Write 1.2 as a fraction in lowest terms.	Simplify 18/15 to lowest terms.	**6/5**
3	Find the remainder r when 92 = 9·q + r, 0 ≤ r < 9.	Find the remainder r when -38 = 5·q + r, 0 ≤ r < 5.	**2**
4	Find gcd(60, 24).	Find gcd(12, 12).	**12**
5	How many positive divisors does 10 have?	How many positive divisors does 34 have?	**4**
6	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $2^n + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, 2n + 3 is prime".	**3**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	Find gcd(40, 16).	Find gcd(8, 48).	**8**
2	Write 0.5 as a fraction in lowest terms.	Simplify 3/6 to lowest terms.	**1/2**
3	Find the remainder r when 36 = 7·q + r, 0 ≤ r < 7.	Find the remainder r when -56 = 3·q + r, 0 ≤ r < 3.	**1**
4	Find gcd(44, 18).	Find gcd(36, 14).	**2**
5	How many positive divisors does 86 have?	How many positive divisors does 35 have?	**4**
6	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $2^n + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $n^2 + 1$ is prime".	**3**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $2^n + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $n^2 + 1$ is prime".	**3**
2	Find gcd(45, 36).	Find gcd(36, 9).	**9**
3	Find the remainder r when 276 = 5·q + r, 0 ≤ r < 5.	Find the remainder r when -29 = 3·q + r, 0 ≤ r < 3.	**1**
4	Write 0.5 as a fraction in lowest terms.	Simplify 3/6 to lowest terms.	**1/2**
5	How many positive divisors does 105 have?	How many positive divisors does 104 have?	**8**
6	Find gcd(20, 18).	Find gcd(24, 14).	**2**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	How many positive divisors does 7 have?	How many positive divisors does 29 have?	**2**
2	Find gcd(44, 21).	Find gcd(36, 35).	**1**
3	Write 1.6 as a fraction in lowest terms.	Simplify 24/15 to lowest terms.	**8/5**
4	Find the remainder r when 119 = 7·q + r, 0 ≤ r < 7.	Find the remainder r when -12 = 4·q + r, 0 ≤ r < 4.	**0**
5	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, $2^n + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all n ≥ 1, 2n + 3 is prime".	**3**
6	Find the remainder r when 52 = 8·q + r, 0 ≤ r < 8.	Find the remainder r when -41 = 5·q + r, 0 ≤ r < 5.	**4**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	Write 0.5 as a fraction in lowest terms.	Simplify 3/6 to lowest terms.	**1/2**
2	Find gcd(55, 60).	Find gcd(55, 40).	**5**
3	Find the remainder r when $240 = 3 \cdot q + r$, $0 \le r < 3$.	Find the remainder r when $-42 = 6 \cdot q + r$, $0 \le r < 6$.	**0**
4	Find the smallest positive integer n that is a counterexample to "for all $n \ge 1$, $2^n + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all $n \ge 1$, $n^2 + 1$ is prime".	**3**
5	How many positive divisors does 106 have?	How many positive divisors does 39 have?	**4**
6	Find the remainder r when $187 = 6 \cdot q + r$, $0 \le r < 6$.	Find the remainder r when $-44 = 9 \cdot q + r$, $0 \le r < 9$.	**1**

Unit 3 · Elementary Number Theory and Methods of Proof · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Find the remainder r when $134 = 7 \cdot q + r$, $0 \le r < 7$.	Find the remainder r when $-8 = 3 \cdot q + r$, $0 \le r < 3$.	**1**
2	Write 0.625 as a fraction in lowest terms.	Simplify 15/24 to lowest terms.	**5/8**
3	Find the remainder r when $342 = 9 \cdot q + r$, $0 \le r < 9$.	Find the remainder r when $-42 = 6 \cdot q + r$, $0 \le r < 6$.	**0**
4	How many positive divisors does 140 have?	How many positive divisors does 72 have?	**12**
5	Find gcd(16, 8).	Find gcd(8, 48).	**8**
6	Find the smallest positive integer n that is a counterexample to "for all $n \ge 1$, $n! + 1$ is prime".	Find the smallest positive integer n that is a counterexample to "for all $n \ge 1$, $n^2 + n + 1$ is prime".	**4**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	Evaluate 7!/6!.	How many ways can 1 different medals be given to 1 of 7 runners?	**7**
2	Evaluate C(4, 2).	How many 2-element subsets does a 4-element set have?	**6**
3	Arithmetic sequence: $a_1 = -3$, $d = 3$. Find a_7.	$a_1 = 0$ and a_n = a_(n−1) + 3. Find a_6.	**15**
4	$a_0 = 3$, a_n = 2a_(n−1) + 1. Find a_5.	How many moves does the Tower of Hanoi need for 7 disks (m_k = 2m_(k−1) + 1, m_1 = 1)?	**127**
5	Evaluate 1 + 2 + ... + 12.	Evaluate Σ_{k=1}^{12} k.	**78**
6	Arithmetic sequence: $a_1 = 8$, $d = 6$. Find a_10.	$a_1 = 2$ and a_n = a_(n−1) + 6. Find a_11.	**62**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	Evaluate 1 + 2 + ... + 13.	Evaluate Σ_{k=1}^{13} k.	**91**
2	Arithmetic sequence: $a_1 = -2$, $d = 4$. Find a_16.	$a_1 = 2$ and a_n = a_(n−1) + 4. Find a_15.	**58**
3	Evaluate C(4, 2).	How many 2-element subsets does a 4-element set have?	**6**
4	Arithmetic sequence: $a_1 = -3$, $d = 4$. Find a_13.	$a_1 = 9$ and a_n = a_(n−1) + 4. Find a_10.	**45**
5	Evaluate 7!/6!.	How many ways can 1 different medals be given to 1 of 7 runners?	**7**
6	$a_0 = 3$, a_n = 2a_(n−1) + 1. Find a_4.	How many moves does the Tower of Hanoi need for 6 disks (m_k = 2m_(k−1) + 1, m_1 = 1)?	**63**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	Evaluate 5!/3!.	How many ways can 2 different medals be given to 2 of 5 runners?	**20**
2	Arithmetic sequence: $a_1 = -2$, $d = 2$. Find a_{16}.	$a_1 = 0$ and $a_n = a_{n-1} + 2$. Find a_{15}.	**28**
3	Evaluate $C(7, 2)$.	How many 2-element subsets does a 7-element set have?	**21**
4	Arithmetic sequence: $a_1 = -1$, $d = 2$. Find a_{15}.	$a_1 = -3$ and $a_n = a_{n-1} + 2$. Find a_{16}.	**27**
5	Evaluate $1 + 2 + \ldots + 18$.	Evaluate $\sum_{k=1}^{18} k$.	**171**
6	$a_0 = 3$, $a_n = 2a_{n-1} + 1$. Find a_5.	How many moves does the Tower of Hanoi need for 7 disks ($m_k = 2m_{k-1} + 1$, $m_1 = 1$)?	**127**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	Arithmetic sequence: $a_1 = -1$, $d = 4$. Find a_7.	$a_1 = -1$ and $a_n = a_{n-1} + 3$. Find a_9.	**23**
2	Evaluate $C(4, 2)$.	How many 2-element subsets does a 4-element set have?	**6**
3	Evaluate 7!/4!.	How many ways can 3 different medals be given to 3 of 7 runners?	**210**
4	Evaluate $C(5, 2)$.	How many 2-element subsets does a 5-element set have?	**10**
5	Evaluate $1 + 2 + \ldots + 11$.	Evaluate $\sum_{k=1}^{11} k$.	**66**
6	$a_0 = 3$, $a_n = 2a_{n-1} + 1$. Find a_5.	How many moves does the Tower of Hanoi need for 7 disks ($m_k = 2m_{k-1} + 1$, $m_1 = 1$)?	**127**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	$a_0 = 3$, a_n = 2a_(n−1) + 1. Find a_3.	How many moves does the Tower of Hanoi need for 5 disks (m_k = 2m_(k−1) + 1, m_1 = 1)?	**31**
2	Evaluate 9!/7!.	How many ways can 2 different medals be given to 2 of 9 runners?	**72**
3	Evaluate C(6, 2).	How many 2-element subsets does a 6-element set have?	**15**
4	Evaluate C(7, 5).	How many 2-element subsets does a 7-element set have?	**21**
5	Evaluate 1 + 2 + ... + 14.	Evaluate Σ_{k=1}^{14} k.	**105**
6	Arithmetic sequence: $a_1 = 4$, d = 2. Find a_13.	$a_1 = 10$ and a_n = a_(n−1) + 3. Find a_7.	**28**

Unit 4 · Sequences, Mathematical Induction, and Recursion · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Evaluate C(7, 5).	How many 2-element subsets does a 7-element set have?	**21**
2	Evaluate 1 + 2 + ... + 20.	Evaluate Σ_{k=1}^{20} k.	**210**
3	Evaluate 6!/5!.	How many ways can 1 different medals be given to 1 of 6 runners?	**6**
4	$a_0 = 1$, a_n = 2a_(n−1) + 1. Find a_5.	How many moves does the Tower of Hanoi need for 6 disks (m_k = 2m_(k−1) + 1, m_1 = 1)?	**63**
5	Arithmetic sequence: $a_1 = 1$, d = 5. Find a_13.	$a_1 = 1$ and a_n = a_(n−1) + 4. Find a_16.	**61**
6	Evaluate C(5, 2).	How many 2-element subsets does a 5-element set have?	**10**

Unit 5 · Set Theory · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	A = {2, 3, 4, 5, 10}, B = {1, 2, 5, 7, 10}. Find A − B.	U = {1, ..., 10}, A = {1, 2, 5, 6, 8}, B = {3, 4, 5, 8}. Find B − A.	**{3, 4}**
2	A = {1, 4, 8, 9, 10}, B = {2, 3, 4, 7, 10}. Find A ∩ B.	U = {1, ..., 10}, A = {5, 6, 7, 8, 9}, B = {4, 8, 9, 10}. Find B − A.	**{4, 10}**
3	How many elements does the power set of a 2-element set have?	How many subsets of {1, ..., 3} contain the element 1?	**4**
4	|A| = 21, |B| = 23, |A ∩ B| = 1. Find |A ∪ B|.	A Venn diagram has 15 in A only, 20 in B only and 8 in both. How many are in A ∪ B?	**43**
5	How many elements does the power set of a 5-element set have?	How many subsets of {1, ..., 6} contain the element 1?	**32**
6	|A| = 17, |B| = 15, |A ∩ B| = 5. Find |A ∪ B|.	A Venn diagram has 11 in A only, 9 in B only and 7 in both. How many are in A ∪ B?	**27**

Unit 5 · Set Theory · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	How many elements does the power set of a 1-element set have?	How many subsets of {1, ..., 2} contain the element 1?	**2**
2	A = {1, 3, 4, 5, 8}, B = {2, 4, 8, 9, 10}. Find A ∩ B.	U = {1, ..., 10}, A = {2, 6, 7, 9, 10}, B = {1, 3, 5, 9}. Find (A ∪ B)'.	**{4, 8}**
3	|A| = 19, |B| = 9, |A ∩ B| = 5. Find |A ∪ B|.	A Venn diagram has 4 in A only, 15 in B only and 4 in both. How many are in A ∪ B?	**23**
4	|A| = 13, |B| = 25, |A ∩ B| = 6. Find |A ∪ B|.	A Venn diagram has 10 in A only, 15 in B only and 7 in both. How many are in A ∪ B?	**32**
5	How many elements does the power set of a 4-element set have?	How many subsets of {1, ..., 5} contain the element 1?	**16**
6	A = {2, 3, 4, 6, 7}, B = {1, 2, 6, 9, 10}. Find A ∩ B.	U = {1, ..., 10}, A = {3, 4, 5, 7, 8}, B = {1, 4, 9, 10}. Find (A ∪ B)'.	**{2, 6}**

Unit 5 · Set Theory · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	A = {1, 2, 3, 5, 7}, B = {4, 5, 6, 7, 9}. Find A − B.	U = {1, ..., 10}, A = {4, 5, 7, 9, 10}, B = {5, 6, 8, 9}. Find (A ∪ B)'.	**{1, 2, 3}**
2	|A| = 11, |B| = 28, |A ∩ B| = 4. Find |A ∪ B|.	A Venn diagram has 13 in A only, 18 in B only and 4 in both. How many are in A ∪ B?	**35**
3	|A| = 12, |B| = 22, |A ∩ B| = 6. Find |A ∪ B|.	A Venn diagram has 15 in A only, 9 in B only and 4 in both. How many are in A ∪ B?	**28**
4	How many elements does the power set of a 1-element set have?	How many subsets of {1, ..., 2} contain the element 1?	**2**
5	How many elements does the power set of a 5-element set have?	How many subsets of {1, ..., 6} contain the element 1?	**32**
6	A = {2, 4, 5, 7, 8}, B = {2, 6, 7, 8, 10}. Find A − B.	U = {1, ..., 10}, A = {2, 6, 8, 9, 10}, B = {1, 3, 6, 7}. Find (A ∪ B)'.	**{4, 5}**

Unit 5 · Set Theory · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	|A| = 26, |B| = 8, |A ∩ B| = 5. Find |A ∪ B|.	A Venn diagram has 6 in A only, 17 in B only and 6 in both. How many are in A ∪ B?	**29**
2	A = {2, 3, 4, 5, 9}, B = {4, 5, 6, 7, 9}. Find A − B.	U = {1, ..., 10}, A = {1, 6, 7, 8, 9}, B = {2, 3, 7, 9}. Find B − A.	**{2, 3}**
3	How many elements does the power set of a 1-element set have?	How many subsets of {1, ..., 2} contain the element 1?	**2**
4	A = {2, 3, 5, 7, 10}, B = {2, 4, 7, 9, 10}. Find A − B.	U = {1, ..., 10}, A = {1, 2, 8, 9, 10}, B = {3, 5, 8, 10}. Find B − A.	**{3, 5}**
5	|A| = 23, |B| = 12, |A ∩ B| = 7. Find |A ∪ B|.	A Venn diagram has 14 in A only, 13 in B only and 1 in both. How many are in A ∪ B?	**28**
6	How many elements does the power set of a 4-element set have?	How many subsets of {1, ..., 5} contain the element 1?	**16**

Unit 5 · Set Theory · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	$\|A\| = 19$, $\|B\| = 20$, $\|A \cap B\| = 7$. Find $\|A \cup B\|$.	A Venn diagram has 10 in A only, 14 in B only and 8 in both. How many are in $A \cup B$?	**32**
2	$A = \{1, 2, 4, 5, 8\}$, $B = \{1, 6, 7, 8, 9\}$. Find $A \cap B$.	$U = \{1, ..., 10\}$, $A = \{2, 3, 5, 6, 7\}$, $B = \{4, 7, 9, 10\}$. Find $(A \cup B)'$.	**{1, 8}**
3	$A = \{5, 7, 8, 9, 10\}$, $B = \{1, 2, 3, 5, 6\}$. Find $A - B$.	$U = \{1, ..., 10\}$, $A = \{1, 2, 4, 5, 6\}$, $B = \{1, 2, 3, 5\}$. Find $(A \cup B)'$.	**{7, 8, 9, 10}**
4	How many elements does the power set of a 2-element set have?	How many subsets of $\{1, ..., 3\}$ contain the element 1?	**4**
5	How many elements does the power set of a 3-element set have?	How many subsets of $\{1, ..., 4\}$ contain the element 1?	**8**
6	$\|A\| = 22$, $\|B\| = 23$, $\|A \cap B\| = 2$. Find $\|A \cup B\|$.	A Venn diagram has 20 in A only, 18 in B only and 5 in both. How many are in $A \cup B$?	**43**

Unit 5 · Set Theory · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	How many elements does the power set of a 4-element set have?	How many subsets of $\{1, ..., 5\}$ contain the element 1?	**16**
2	$A = \{1, 2, 5, 7, 9\}$, $B = \{1, 3, 6, 8, 9\}$. Find $A - B$.	$U = \{1, ..., 10\}$, $A = \{1, 4, 6, 8, 10\}$, $B = \{3, 4, 8, 9\}$. Find $(A \cup B)'$.	**{2, 5, 7}**
3	How many elements does the power set of a 1-element set have?	How many subsets of $\{1, ..., 2\}$ contain the element 1?	**2**
4	$\|A\| = 23$, $\|B\| = 8$, $\|A \cap B\| = 6$. Find $\|A \cup B\|$.	A Venn diagram has 8 in A only, 16 in B only and 1 in both. How many are in $A \cup B$?	**25**
5	$A = \{1, 5, 6, 9, 10\}$, $B = \{2, 3, 4, 6, 8\}$. Find $A - B$.	$U = \{1, ..., 10\}$, $A = \{2, 3, 4, 6, 8\}$, $B = \{3, 4, 7, 8\}$. Find $(A \cup B)'$.	**{1, 5, 9, 10}**
6	$\|A\| = 27$, $\|B\| = 12$, $\|A \cap B\| = 1$. Find $\|A \cup B\|$.	A Venn diagram has 15 in A only, 18 in B only and 5 in both. How many are in $A \cup B$?	**38**

Unit 6 · Properties of Relations · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	How many ordered pairs are in "a < b" on {1, ..., 3}?	How many ordered pairs are in "a > b" on {1, ..., 3}?	**3**
2	A partition of a set has blocks of sizes 4, 2. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 2, 4. How many ordered pairs are in the induced relation?	**20**
3	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
4	How many ordered pairs are in "a < b" on {1, ..., 4}?	How many ordered pairs are in "a > b" on {1, ..., 4}?	**6**
5	Congruence mod 4 on {0, 1, ..., 12}. Find [4].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 4. List all b with 4 R b.	**{0, 4, 8, 12}**
6	Congruence mod 5 on {0, 1, ..., 12}. Find [4].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 4 R b.	**{4, 9}**

Unit 6 · Properties of Relations · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	Congruence mod 5 on {0, 1, ..., 12}. Find [4].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 4 R b.	**{4, 9}**
2	How many ordered pairs are in "a < b" on {1, ..., 8}?	How many ordered pairs are in "a > b" on {1, ..., 8}?	**28**
3	Congruence mod 4 on {0, 1, ..., 12}. Find [3].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 4. List all b with 3 R b.	**{3, 7, 11}**
4	How many ordered pairs are in "a < b" on {1, ..., 3}?	How many ordered pairs are in "a > b" on {1, ..., 3}?	**3**
5	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
6	A partition of a set has blocks of sizes 3, 4, 4. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 4, 4, 3. How many ordered pairs are in the induced relation?	**41**

Unit 6 · Properties of Relations · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
2	How many ordered pairs are in "a < b" on {1, ..., 5}?	How many ordered pairs are in "a > b" on {1, ..., 5}?	**10**
3	How many ordered pairs are in "a < b" on {1, ..., 8}?	How many ordered pairs are in "a > b" on {1, ..., 8}?	**28**
4	Congruence mod 5 on {0, 1, ..., 12}. Find [1].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 1 R b.	**{1, 6, 11}**
5	Congruence mod 5 on {0, 1, ..., 12}. Find [2].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 2 R b.	**{2, 7, 12}**
6	A partition of a set has blocks of sizes 4, 4, 1. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 4, 1, 4. How many ordered pairs are in the induced relation?	**33**

Unit 6 · Properties of Relations · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	A partition of a set has blocks of sizes 2, 1, 2. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 1, 2, 2. How many ordered pairs are in the induced relation?	**9**
2	Congruence mod 5 on {0, 1, ..., 12}. Find [4].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 4 R b.	**{4, 9}**
3	How many ordered pairs are in "a < b" on {1, ..., 5}?	How many ordered pairs are in "a > b" on {1, ..., 5}?	**10**
4	Congruence mod 4 on {0, 1, ..., 12}. Find [3].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 4. List all b with 3 R b.	**{3, 7, 11}**
5	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
6	A partition of a set has blocks of sizes 3, 2. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 2, 3. How many ordered pairs are in the induced relation?	**13**

Unit 6 · Properties of Relations · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	A partition of a set has blocks of sizes 4, 1, 2. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 2, 4, 1. How many ordered pairs are in the induced relation?	**21**
2	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
3	How many ordered pairs are in "a < b" on {1, ..., 5}?	How many ordered pairs are in "a > b" on {1, ..., 5}?	**10**
4	Congruence mod 4 on {0, 1, ..., 12}. Find [4].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 4. List all b with 0 R b.	**{0, 4, 8, 12}**
5	A partition of a set has blocks of sizes 1, 2. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 2, 1. How many ordered pairs are in the induced relation?	**5**
6	Congruence mod 5 on {0, 1, ..., 12}. Find [0].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 5. List all b with 0 R b.	**{0, 5, 10}**

Unit 6 · Properties of Relations · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	A partition of a set has blocks of sizes 2, 4, 4. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 4, 4, 2. How many ordered pairs are in the induced relation?	**36**
2	Congruence mod 3 on {0, 1, ..., 12}. Find [3].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 3. List all b with 0 R b.	**{0, 3, 6, 9, 12}**
3	How many ordered pairs are in "a < b" on {1, ..., 3}?	How many ordered pairs are in "a > b" on {1, ..., 3}?	**3**
4	Congruence mod 3 on {0, 1, ..., 12}. Find [1].	On {0, ..., 12}, a R b ⟺ a and b have the same remainder mod 3. List all b with 4 R b.	**{1, 4, 7, 10}**
5	How many reflexive relations are there on a 3-element set?	How many symmetric relations are there on a 3-element set?	**64**
6	A partition of a set has blocks of sizes 2, 1. How many ordered pairs are in the induced relation?	A partition of a set has blocks of sizes 1, 2. How many ordered pairs are in the induced relation?	**5**

Unit 7 · Counting and Probability · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	How many integers are in the list 15, 16, ..., 50?	How many integers are in the list 22, 23, ..., 57?	**36**
2	3 shirts, 2 pants and 4 hats: how many outfits (one of each)?	6 shirts, 4 pants and 1 hats: how many outfits (one of each)?	**24**
3	Two fair dice: find P(sum = 9).	A bag has 2 red and 16 blue marbles. Find P(red).	**1/9**
4	5 shirts, 2 pants and 2 hats: how many outfits (one of each)?	4 shirts, 5 pants and 1 hats: how many outfits (one of each)?	**20**
5	How many 4-person committees can be chosen from 4 people?	How many bit strings of length 4 have exactly 4 ones?	**1**
6	How many ways can 2 of 4 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 4-letter alphabet?	**12**

Unit 7 · Counting and Probability · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	How many ways can 3 of 5 people fill 3 different offices?	How many 3-letter strings with no repeated letter use letters from a 5-letter alphabet?	**60**
2	Two fair dice: find P(sum = 6).	A bag has 5 red and 31 blue marbles. Find P(red).	**5/36**
3	5 shirts, 2 pants and 1 hats: how many outfits (one of each)?	2 shirts, 5 pants and 1 hats: how many outfits (one of each)?	**10**
4	How many integers are in the list 28, 29, ..., 71?	How many integers are in the list 14, 15, ..., 57?	**44**
5	5 shirts, 6 pants and 3 hats: how many outfits (one of each)?	6 shirts, 5 pants and 3 hats: how many outfits (one of each)?	**90**
6	How many 3-person committees can be chosen from 4 people?	How many bit strings of length 4 have exactly 3 ones?	**4**

Unit 7 · Counting and Probability · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	How many 2-person committees can be chosen from 7 people?	How many bit strings of length 7 have exactly 2 ones?	**21**
2	5 shirts, 4 pants and 3 hats: how many outfits (one of each)?	5 shirts, 6 pants and 2 hats: how many outfits (one of each)?	**60**
3	2 shirts, 4 pants and 3 hats: how many outfits (one of each)?	3 shirts, 4 pants and 2 hats: how many outfits (one of each)?	**24**
4	How many ways can 3 of 7 people fill 3 different offices?	How many 3-letter strings with no repeated letter use letters from a 7-letter alphabet?	**210**
5	How many integers are in the list 6, 7, ..., 83?	How many integers are in the list 21, 22, ..., 98?	**78**
6	Two fair dice: find P(sum = 3).	A bag has 2 red and 34 blue marbles. Find P(red).	**1/18**

Unit 7 · Counting and Probability · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	How many 2-person committees can be chosen from 4 people?	How many bit strings of length 4 have exactly 2 ones?	**6**
2	Two fair dice: find P(sum = 4).	A bag has 3 red and 33 blue marbles. Find P(red).	**1/12**
3	2 shirts, 4 pants and 4 hats: how many outfits (one of each)?	4 shirts, 4 pants and 2 hats: how many outfits (one of each)?	**32**
4	How many ways can 2 of 6 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 6-letter alphabet?	**30**
5	How many ways can 2 of 8 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 8-letter alphabet?	**56**
6	How many integers are in the list 37, 38, ..., 104?	How many integers are in the list 12, 13, ..., 79?	**68**

Unit 7 · Counting and Probability · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	2 shirts, 4 pants and 2 hats: how many outfits (one of each)?	4 shirts, 2 pants and 2 hats: how many outfits (one of each)?	**16**
2	Two fair dice: find P(sum = 8).	A bag has 5 red and 31 blue marbles. Find P(red).	**5/36**
3	How many ways can 2 of 8 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 8-letter alphabet?	**56**
4	How many 2-person committees can be chosen from 7 people?	How many bit strings of length 7 have exactly 2 ones?	**21**
5	How many integers are in the list 31, 32, ..., 116?	How many integers are in the list 34, 35, ..., 119?	**86**
6	How many ways can 3 of 4 people fill 3 different offices?	How many 3-letter strings with no repeated letter use letters from a 4-letter alphabet?	**24**

Unit 7 · Counting and Probability · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	How many ways can 2 of 5 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 5-letter alphabet?	**20**
2	How many 2-person committees can be chosen from 8 people?	How many bit strings of length 8 have exactly 2 ones?	**28**
3	5 shirts, 2 pants and 1 hats: how many outfits (one of each)?	2 shirts, 5 pants and 1 hats: how many outfits (one of each)?	**10**
4	How many integers are in the list 11, 12, ..., 61?	How many integers are in the list 8, 9, ..., 58?	**51**
5	How many ways can 2 of 4 people fill 2 different offices?	How many 2-letter strings with no repeated letter use letters from a 4-letter alphabet?	**12**
6	Two fair dice: find P(sum = 11).	A bag has 1 red and 17 blue marbles. Find P(red).	**1/18**

Unit 8 · Theory of Graphs and Trees · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	How many edges does K_6 have?	At a party, each of 6 people shakes hands once with every other person. How many handshakes?	**15**
2	What is the chromatic number of the cycle C_7?	What is the chromatic number of K_3?	**3**
3	How many edges does K_3,3 have?	How many edges does a Hamiltonian cycle in a graph with 9 vertices use?	**9**
4	A graph has degrees 5, 5, 4, 5, 1. How many edges?	Every vertex of a graph with 5 vertices has degree 4. How many edges?	**10**
5	How many edges does a tree with 9 vertices have?	A graph has degree sum 16. How many edges does it have?	**8**
6	How many edges does K_8 have?	At a party, each of 8 people shakes hands once with every other person. How many handshakes?	**28**

Unit 8 · Theory of Graphs and Trees · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	How many edges does K_6 have?	At a party, each of 6 people shakes hands once with every other person. How many handshakes?	**15**
2	A graph has degrees 6, 2, 1, 5, 1, 1. How many edges?	Every vertex of a graph with 8 vertices has degree 2. How many edges?	**8**
3	How many edges does a tree with 5 vertices have?	A graph has degree sum 8. How many edges does it have?	**4**
4	How many edges does K_2,5 have?	How many edges does a Hamiltonian cycle in a graph with 10 vertices use?	**10**
5	What is the chromatic number of the cycle C_5?	What is the chromatic number of K_3?	**3**
6	How many edges does K_8 have?	At a party, each of 8 people shakes hands once with every other person. How many handshakes?	**28**

Unit 8 · Theory of Graphs and Trees · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	How many edges does a tree with 11 vertices have?	A graph has degree sum 20. How many edges does it have?	**10**
2	A graph has degrees 1, 2, 1, 5, 1. How many edges?	Every vertex of a graph with 5 vertices has degree 2. How many edges?	**5**
3	How many edges does K_4 have?	At a party, each of 4 people shakes hands once with every other person. How many handshakes?	**6**
4	How many edges does K_2,2 have?	How many edges does a Hamiltonian cycle in a graph with 4 vertices use?	**4**
5	What is the chromatic number of the cycle C_3?	What is the chromatic number of K_3?	**3**
6	How many edges does K_6 have?	At a party, each of 6 people shakes hands once with every other person. How many handshakes?	**15**

Unit 8 · Theory of Graphs and Trees · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	How many edges does K_9 have?	At a party, each of 9 people shakes hands once with every other person. How many handshakes?	**36**
2	What is the chromatic number of the cycle C_5?	What is the chromatic number of K_3?	**3**
3	How many edges does a tree with 18 vertices have?	A graph has degree sum 34. How many edges does it have?	**17**
4	How many edges does a tree with 11 vertices have?	A graph has degree sum 20. How many edges does it have?	**10**
5	How many edges does K_2,6 have?	How many edges does a Hamiltonian cycle in a graph with 12 vertices use?	**12**
6	A graph has degrees 5, 3, 1, 3. How many edges?	Every vertex of a graph with 6 vertices has degree 2. How many edges?	**6**

Unit 8 · Theory of Graphs and Trees · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	How many edges does a tree with 11 vertices have?	A graph has degree sum 20. How many edges does it have?	**10**
2	What is the chromatic number of the cycle C_7?	What is the chromatic number of K_3?	**3**
3	How many edges does $K_{4,3}$ have?	How many edges does a Hamiltonian cycle in a graph with 12 vertices use?	**12**
4	How many edges does K_6 have?	At a party, each of 6 people shakes hands once with every other person. How many handshakes?	**15**
5	A graph has degrees 4, 4, 1, 1, 2. How many edges?	Every vertex of a graph with 6 vertices has degree 2. How many edges?	**6**
6	How many edges does a tree with 17 vertices have?	A graph has degree sum 32. How many edges does it have?	**16**

Unit 8 · Theory of Graphs and Trees · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	What is the chromatic number of the cycle C_6?	What is the chromatic number of K_2?	**2**
2	How many edges does a tree with 14 vertices have?	A graph has degree sum 26. How many edges does it have?	**13**
3	How many edges does a tree with 20 vertices have?	A graph has degree sum 38. How many edges does it have?	**19**
4	How many edges does $K_{3,4}$ have?	How many edges does a Hamiltonian cycle in a graph with 12 vertices use?	**12**
5	A graph has degrees 3, 2, 2, 1. How many edges?	Every vertex of a graph with 4 vertices has degree 2. How many edges?	**4**
6	How many edges does K_4 have?	At a party, each of 4 people shakes hands once with every other person. How many handshakes?	**6**

Unit 9 · Election Theory · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	How many votes are a majority of 21 votes?	Weights 6, 6, 2, 6: what is the smallest quota that is more than half the total?	**11**
2	Total population 2790, 9 seats. Find the standard divisor.	Total population 1550, 5 seats. Find the standard divisor.	**310**
3	How many votes are a majority of 23 votes?	Weights 8, 6, 9: what is the smallest quota that is more than half the total?	**12**
4	5 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	5 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**20**
5	7 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	7 voters rank X second among 4 candidates (Borda: n points for 1st). How many points does X get from them?	**21**
6	In [8: 8, 4, 2], how many times is P1 critical (over all winning coalitions)?	In [7: 8, 2, 2], how many times is P1 critical (over all winning coalitions)?	**4**

Unit 9 · Election Theory · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	12 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	12 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**48**
2	How many votes are a majority of 30 votes?	Weights 7, 9, 6, 9: what is the smallest quota that is more than half the total?	**16**
3	In [6: 4, 4, 3], how many times is P1 critical (over all winning coalitions)?	In [10: 8, 6, 5], how many times is P1 critical (over all winning coalitions)?	**2**
4	Total population 6720, 14 seats. Find the standard divisor.	Total population 2880, 6 seats. Find the standard divisor.	**480**
5	How many votes are a majority of 28 votes?	Weights 7, 7, 8, 6: what is the smallest quota that is more than half the total?	**15**
6	8 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	8 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**32**

Unit 9 · Election Theory · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	In [8: 6, 5, 4], how many times is P1 critical (over all winning coalitions)?	In [11: 8, 8, 5], how many times is P1 critical (over all winning coalitions)?	**2**
2	How many votes are a majority of 26 votes?	Weights 9, 8, 6, 4: what is the smallest quota that is more than half the total?	**14**
3	Total population 1800, 12 seats. Find the standard divisor.	Total population 3150, 21 seats. Find the standard divisor.	**150**
4	6 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	6 voters rank X second among 4 candidates (Borda: n points for 1st). How many points does X get from them?	**18**
5	How many votes are a majority of 31 votes?	Weights 9, 9, 8, 4: what is the smallest quota that is more than half the total?	**16**
6	11 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	11 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**44**

Unit 9 · Election Theory · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	Total population 12320, 14 seats. Find the standard divisor.	Total population 15840, 18 seats. Find the standard divisor.	**880**
2	How many votes are a majority of 19 votes?	Weights 4, 4, 9, 2: what is the smallest quota that is more than half the total?	**10**
3	In [6: 8, 1, 1], how many times is P1 critical (over all winning coalitions)?	In [6: 7, 3, 1], how many times is P1 critical (over all winning coalitions)?	**4**
4	4 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	3 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**12**
5	5 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	5 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**20**
6	In [8: 7, 5, 3], how many times is P1 critical (over all winning coalitions)?	In [10: 7, 6, 5], how many times is P1 critical (over all winning coalitions)?	**2**

Unit 9 · Election Theory · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	How many votes are a majority of 14 votes?	Weights 4, 3, 1, 6: what is the smallest quota that is more than half the total?	**8**
2	In [9: 6, 5, 5], how many times is P1 critical (over all winning coalitions)?	In [12: 8, 7, 7], how many times is P1 critical (over all winning coalitions)?	**2**
3	In [8: 8, 4, 2], how many times is P1 critical (over all winning coalitions)?	In [6: 6, 3, 1], how many times is P1 critical (over all winning coalitions)?	**4**
4	9 voters rank X first among 4 candidates (Borda: n points for 1st). How many points does X get from them?	9 voters rank X second among 5 candidates (Borda: n points for 1st). How many points does X get from them?	**36**
5	Total population 10880, 17 seats. Find the standard divisor.	Total population 13440, 21 seats. Find the standard divisor.	**640**
6	3 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	3 voters rank X second among 4 candidates (Borda: n points for 1st). How many points does X get from them?	**9**

Unit 9 · Election Theory · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Total population 10350, 15 seats. Find the standard divisor.	Total population 13110, 19 seats. Find the standard divisor.	**690**
2	8 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	8 voters rank X second among 4 candidates (Borda: n points for 1st). How many points does X get from them?	**24**
3	In [7: 5, 5, 3], how many times is P1 critical (over all winning coalitions)?	In [9: 8, 6, 3], how many times is P1 critical (over all winning coalitions)?	**2**
4	10 voters rank X first among 3 candidates (Borda: n points for 1st). How many points does X get from them?	10 voters rank X second among 4 candidates (Borda: n points for 1st). How many points does X get from them?	**30**
5	How many votes are a majority of 24 votes?	Weights 9, 8, 7, 1: what is the smallest quota that is more than half the total?	**13**
6	In [6: 7, 3, 1], how many times is P1 critical (over all winning coalitions)?	In [7: 7, 4, 1], how many times is P1 critical (over all winning coalitions)?	**4**

Unit 10 · Linear Programming · Partner Check 1 · CP2

#	Partner A	Partner B	Answer
1	Maximize P = 4x + 4y subject to 2x + 3y ≤ 17, 3x + y ≤ 15, x, y ≥ 0. Give the maximum value.	Maximize P = 4x + 4y subject to 2x + 3y ≤ 14, x + 2y ≤ 11, x, y ≥ 0. Give the maximum value.	**28**
2	Evaluate P = 1x + 7y at (8, 1).	Evaluate P = 1x + 2y at (1, 7).	**15**
3	At (1, 1), how much slack does 3x + 2y ≤ 34 have?	At (1, 0), how much slack does 3x + 1y ≤ 32 have?	**29**
4	Find the corner where 2x + 2y = 20 meets 1x + 2y = 15.	Find the corner where 1x + 2y = 15 meets 1x + 3y = 20.	**(5, 5)**
5	Find the corner where 3x + 1y = 9 meets 2x + 2y = 10.	Find the corner where 2x + 3y = 13 meets 3x + 3y = 15.	**(2, 3)**
6	Maximize P = 2x + 4y subject to x + 2y ≤ 8, 3x + 2y ≤ 15, x, y ≥ 0. Give the maximum value.	Maximize P = 3x + 1y subject to 2x + 2y ≤ 16, 3x + 3y ≤ 16, x, y ≥ 0. Give the maximum value.	**16**

Unit 10 · Linear Programming · Partner Check 1 · CP1

#	Partner A	Partner B	Answer
1	Maximize P = 1x + 4y subject to 2x + 2y ≤ 8, x + 2y ≤ 8, x, y ≥ 0. Give the maximum value.	Maximize P = 2x + 3y subject to x + 2y ≤ 16, 2x + 3y ≤ 16, x, y ≥ 0. Give the maximum value.	**16**
2	Find the corner where 2x + 1y = 14 meets 3x + 2y = 24.	Find the corner where 2x + 3y = 26 meets 1x + 3y = 22.	**(4, 6)**
3	At (2, 3), how much slack does 3x + 1y ≤ 40 have?	At (0, 1), how much slack does 3x + 1y ≤ 32 have?	**31**
4	Maximize P = 3x + 2y subject to x + 2y ≤ 15, 3x + 3y ≤ 17, x, y ≥ 0. Give the maximum value.	Maximize P = 2x + 3y subject to 3x + y ≤ 17, 3x + 3y ≤ 17, x, y ≥ 0. Give the maximum value.	**17**
5	Find the corner where 1x + 2y = 12 meets 1x + 1y = 7.	Find the corner where 1x + 3y = 17 meets 2x + 3y = 19.	**(2, 5)**
6	Evaluate P = 4x + 9y at (0, 4).	Evaluate P = 5x + 3y at (6, 2).	**36**

Unit 10 · Linear Programming · Partner Check 1 · Honors

#	Partner A	Partner B	Answer
1	At (2, 2), how much slack does 1x + 2y ≤ 33 have?	At (0, 1), how much slack does 4x + 4y ≤ 31 have?	**27**
2	Evaluate P = 9x + 4y at (0, 0).	Evaluate P = 1x + 9y at (0, 0).	**0**
3	Find the corner where 3x + 3y = 9 meets 3x + 2y = 7.	Find the corner where 3x + 2y = 7 meets 1x + 1y = 3.	**(1, 2)**
4	Find the corner where 1x + 2y = 8 meets 2x + 1y = 13.	Find the corner where 2x + 3y = 15 meets 2x + 2y = 14.	**(6, 1)**
5	Maximize P = 4x + 2y subject to x + 3y ≤ 6, 2x + y ≤ 18, x, y ≥ 0. Give the maximum value.	Maximize P = 1x + 2y subject to x + y ≤ 12, x + y ≤ 17, x, y ≥ 0. Give the maximum value.	**24**
6	Maximize P = 2x + 4y subject to x + 2y ≤ 18, x + 2y ≤ 7, x, y ≥ 0. Give the maximum value.	Maximize P = 2x + 3y subject to 2x + y ≤ 16, x + 2y ≤ 7, x, y ≥ 0. Give the maximum value.	**14**

Unit 10 · Linear Programming · Partner Check 2 · CP2

#	Partner A	Partner B	Answer
1	At (4, 3), how much slack does 1x + 4y ≤ 33 have?	At (2, 4), how much slack does 2x + 3y ≤ 33 have?	**17**
2	Evaluate P = 9x + 5y at (2, 1).	Evaluate P = 1x + 5y at (8, 3).	**23**
3	Evaluate P = 6x + 7y at (7, 1).	Evaluate P = 6x + 5y at (4, 5).	**49**
4	Find the corner where 2x + 1y = 17 meets 2x + 3y = 27.	Find the corner where 2x + 3y = 27 meets 2x + 2y = 22.	**(6, 5)**
5	Find the corner where 1x + 2y = 7 meets 3x + 3y = 12.	Find the corner where 2x + 3y = 11 meets 1x + 1y = 4.	**(1, 3)**
6	Maximize P = 5x + 4y subject to x + y ≤ 8, x + y ≤ 18, x, y ≥ 0. Give the maximum value.	Maximize P = 5x + 5y subject to 3x + 2y ≤ 17, 3x + y ≤ 8, x, y ≥ 0. Give the maximum value.	**40**

Unit 10 · Linear Programming · Partner Check 2 · CP1

#	Partner A	Partner B	Answer
1	Find the corner where 3x + 2y = 11 meets 3x + 3y = 12.	Find the corner where 2x + 2y = 8 meets 1x + 2y = 5.	**(3, 1)**
2	Maximize P = 4x + 3y subject to 2x + 2y ≤ 6, x + 2y ≤ 10, x, y ≥ 0. Give the maximum value.	Maximize P = 2x + 3y subject to x + 3y ≤ 6, x + 2y ≤ 7, x, y ≥ 0. Give the maximum value.	**12**
3	Find the corner where 3x + 1y = 24 meets 1x + 3y = 24.	Find the corner where 2x + 3y = 30 meets 1x + 3y = 24.	**(6, 6)**
4	At (1, 4), how much slack does 4x + 2y ≤ 39 have?	At (0, 2), how much slack does 4x + 4y ≤ 35 have?	**27**
5	Evaluate P = 4x + 5y at (1, 3).	Evaluate P = 7x + 1y at (2, 5).	**19**
6	Evaluate P = 5x + 9y at (8, 2).	Evaluate P = 4x + 6y at (4, 7).	**58**

Unit 10 · Linear Programming · Partner Check 2 · Honors

#	Partner A	Partner B	Answer
1	Evaluate P = 4x + 2y at (5, 7).	Evaluate P = 7x + 1y at (4, 6).	**34**
2	Find the corner where 3x + 2y = 11 meets 1x + 3y = 13.	Find the corner where 3x + 1y = 7 meets 2x + 3y = 14.	**(1, 4)**
3	Evaluate P = 3x + 7y at (2, 6).	Evaluate P = 1x + 6y at (6, 7).	**48**
4	Maximize P = 4x + 5y subject to x + 2y ≤ 7, 2x + y ≤ 13, x, y ≥ 0. Give the maximum value.	Maximize P = 5x + 4y subject to 3x + 3y ≤ 18, 3x + 2y ≤ 15, x, y ≥ 0. Give the maximum value.	**27**
5	At (3, 2), how much slack does 2x + 2y ≤ 40 have?	At (2, 2), how much slack does 2x + 3y ≤ 40 have?	**30**
6	Find the corner where 3x + 1y = 23 meets 2x + 2y = 22.	Find the corner where 3x + 1y = 23 meets 1x + 1y = 11.	**(6, 5)**